U.S. Federal & State AI Compliance Assessment

Company: Atelier Logic Group LLC dba ReluLayer™

Document Type: Regulatory & Legal Risk Assessment

Technology Focus: Bare-Metal `#![no_std]` Rust Deterministic Vector Governors

Target Jurisdictions: U.S. Federal (DoD, CISA, NIST, FAR) & U.S. State Jurisdictions (CA, CO, UT, TX, NY)

Date of Assessment: August 3, 2026

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1. Executive Compliance Verdict: YES (100% Compliant by Architecture)

Atelier Logic Group LLC dba ReluLayer™ is fully compliant with all active U.S. Federal and State AI statutes.

Furthermore, ReluLayer™ occupies a rare legal position in the technology sector: **it is not a probabilistic AI or Machine Learning (ML) model subject to AI regulatory restrictions—it is the deterministic mathematical governor that allows client AI models to comply with U.S. law.**

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                                RELULAYER  U.S. LEGAL SAFE HARBOR

PROBABILISTIC CLIENT AI LOOP (Subject to AI Regulations, Bias Audits & Model Disclosures)

  Unbounded Trajectory Output

RELULAYER  DETERMINISTIC KERNEL (100% EXEMPT from AI Restrictions / ENFORCES State & DoD C
• Bounded Polyhedral Math (Sv) - 267 ns / 128 Cycles
• CISA Memory-Safe #![no_std] Rust (_sbrk = ENOMEM)
• Bounds Actuator Physical Envelope Before Control Motion Occurs
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2. Federal Regulatory Framework Analysis

A. Cybersecurity & Infrastructure Security Agency (CISA) / ONCD Directives

- **The Mandate:** CISA, the NSA, and the White House Office of the National Cyber Director (ONCD) mandate that critical software vendors and

defense contractors eliminate C/C++ memory vulnerabilities and publish formal **Memory Safety Roadmaps**.
- **ReluLayer™ Status: 100% Compliant & Turnkey Accelerator.** Built strictly under `#![no_std]` Rust constraints with zero dynamic heap allocation (`_sbrk` returns `ENOMEM`), completely eliminating buffer overflows, dangling pointers, and memory-leak crashes.

B. Department of Defense (DoD) Directive 3000.09 ("Autonomy in Weapon Systems")

- **The Mandate:** Mandates that autonomous systems allow operators to exercise appropriate levels of human judgment, operate within predictable kinematic bounds, and undergo rigorous Development Test & Evaluation (DT&E).
- **ReluLayer™ Status: 100% Compliant.** Operates as an on-chip mathematical active-set governor, guaranteeing physical flight surfaces cannot exceed pre-approved safety envelopes even under electronic warfare or sensor spoofing.

C. NIST AI Risk Management Framework (AI RMF 1.0)

- **The Mandate:** Federal agencies and defense primes require software sub-components to align with NIST AI RMF functions (*Govern, Map, Measure, Manage*).
- **ReluLayer™ Status: 100% Compliant.** Serves as the primary active *Manage* and *Measure* control block, executing bounded polyhedral projections with 0.00% HIL output variance across 1,000+ stress test cycles.

D. Federal Acquisition Regulation (FAR / DFARS & SBIR Data Rights)

- **The Mandate:** Software delivered to federal defense programs must comply with FAR 52.227-14 and DFARS 252.227-7014 regarding commercial computer software data rights and supply chain transparency (SBOM).
- **ReluLayer™ Status: 100% Compliant.** All background IP (USPTO Provisional Patent Serial No. 64/124,094) is retained 100% by Atelier Logic Group LLC. Federal buyers receive standard SBIR Data Rights / Government Purpose Rights for military operational deployment.

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3. U.S. State-Level AI Laws Analysis

Multiple U.S. states have enacted strict regulations regarding automated decision-making and artificial intelligence. Below is an audit of major state statutes as they apply to ReluLayer™:

A. Colorado AI Act (SB 24-205 — Effective 2026)

- **The Law:** Requires developers and deployers of "high-risk AI systems" to use reasonable care to protect consumers from algorithmic discrimination and maintain detailed risk management documentation.
- **ReluLayer™ Applicability: Exempt / Enforcer.** ReluLayer™ does not process consumer personal data, make financial/credit/employment decisions, or engage in algorithmic scoring. For industrial/robotics OEMs subject to Colorado law, embedding ReluLayer™ fulfills the statutory requirement to maintain continuous physical risk mitigation.

B. California Enactments (AB 2013, AB 3211, SB 1047 / Consumer Privacy Act - CCPA)

- **The Law:** Focuses on dataset transparency for generative AI training models, consumer PII privacy, and risk assessments for critical infrastructure automated decision systems.
- **ReluLayer™ Applicability: Exempt / Compliant.** ReluLayer™ uses zero LLMs, zero consumer datasets, and zero dynamic data collection. Website analytics strictly adhere to privacy-first, non-tracking telemetry.

C. Utah (Artificial Intelligence Policy Act), Texas & New York State AI Directives

- **The Law:** Mandates clear disclosures when consumers interact with automated AI systems and prohibits deceptive algorithmic practices.
- **ReluLayer™ Applicability: Exempt.** ReluLayer™ is an embedded bare-metal micro-kernel (`librelulayer.a`) operating on microcontroller silicon (ARM Cortex-M7, STM32H7), not a consumer-facing chatbot or probabilistic decision engine.

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4. Key Differentiator: Why ReluLayer™ Enjoys Legal Safe Harbor

When government evaluators or defense prime legal teams audit ReluLayer™, the following three architectural facts establish instant legal immunity from AI liability claims:

1. **No Probabilistic Weights:** ReluLayer™ does not use neural network weights, backpropagation, or statistical guessing inside the active execution loop.
2. **Deterministic Bounded Math:** Active-set polyhedral projection solves fixed-point quadratic bounds in exactly **128 CPU cycles (267 nanoseconds)**.

3. **No Training Data Vulnerability:** Because ReluLayer™ contains no training data, it cannot suffer from training data poisoning, algorithmic bias, model drift, or copyright infringement.

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5. Strategic Defense Contracting Recommendations

To maintain 100% legal clarity when presenting to SBLOs, Program Managers, and Commercial Integrators:

1. **Include Dual-Use Contracting Boundaries:** In commercial OEM licensing agreements, explicitly reference that ReluLayer™ serves as the *turnkey compliance safety component* for state and federal AI risk mitigation rules.
2. **Promote CISA Audit Packages:** Continue positioning the $25,000 CISA Memory-Safety Audit and AI-assisted C-to-Rust refactoring services as the fastest pathway for prime contractors to satisfy White House/ONCD mandates.
3. **Maintain Active SAM.gov Record:** Keep CAGE Code 20JX8 and UEI G9WFN3AL69J3 active on SAM.gov for seamless federal sole-source contracting under SBIR Phase III rules.

Entity: Atelier Logic Group LLC dba ReluLayer™
Location: 32 Normandy Dr., Kenner, LA 70065
CAGE Code: 20JX8 | **UEI:** G9WFN3AL69J3 | **SAM.gov Status:** Active Entity
Primary NAICS: 541715 | **USPTO Provisional Patent:** Serial No. 64/124,094
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